9.7: Maximum Likelihood Estimation

Idea: Choose as an estimate for a parameter to be the value that maximizes the likelihood of the observed sample.

Illustration: Consider a $Binomial(n, p)$ random variable. Suppose it was observed in an experiment that $Y = y$. The maximum likelihood estimator for p would be the value that maximizes $P(Y = y)$.

Definition 9.4 : Likelihood Function

Consider $y_1, \ldots, y_n$ to be a sample of observations taken from the corresponding independent random variables $Y_1, \ldots, Y_n$ whose distribution depends on the parameter θ.

The likelihood function, $L(y_1, \ldots, y_n \mid \theta) = L(\theta)$, is defined as the joint distribution of $Y_1, \ldots, Y_n$ observed in the sample.

$$\Rightarrow L(y_1, \ldots, y_n \mid \theta) = L(\theta) = \begin{cases} p(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} p(y_i \mid \theta), & \text{for d.r.v.'s} \\ f(y_1, \ldots, y_n \mid \theta) = \prod_{i=1}^{n} f(y_i \mid \theta), & \text{for c.r.v.'s} \end{cases}.$$

Method of Maximum Likelihood Estimation :

Choose as estimates for $\theta = (\theta_1, \theta_2, \ldots, \theta_t)$, the values that maximize $L(\theta)$.

Let $\hat{\theta}_{MLE}$ denote the Method of Maximum Likelihood Estimator for θ.

<u>Note:</u> It is often easier to maximize $\ln(L(\theta)) \Rightarrow \ell(\theta)$.

Find the Maximum Likelihood Estimators

Example 4: $Y_1,\ldots,Y_n \overset{ind.}{\sim} Poisson(\lambda)$.

Let $y_1, \ldots y_n$ sample dist of Y_i

$$P(Y_i = y_i) = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}, \quad L(y_1 \cdots y_n | \lambda) = \prod_{i=1}^{n} \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}$$

$$\ln(L(y_1 \cdots y_n | \theta) = \sum_{i=1}^{n} \ln\left(\frac{\lambda^{y_i} e^{-\lambda}}{y_i!}\right) = \sum_{i=1}^{n} \ln(\lambda^{y_i}) + \ln(e^{-\lambda}) - \ln(y_i!)$$

$$= \sum_{i=1}^{n} y_i \ln(\lambda) - \lambda \ln(e) - \ln(y_i!)$$

$$= \sum_{i=1}^{n} y_i \ln(\lambda) - \lambda - \ln(y_i!)$$

$$\therefore \sum_{i=1}^{n} y_i \ln(\lambda) - \lambda - \ln(y_i!)$$

$$= \ln\lambda \left(\sum_{i=1}^{n} y_i\right) - n\lambda - \sum_{i=1}^{n} \ln(y_i!)$$

Compute Derivative wrt to the parameter

$$\frac{1}{\lambda} \cdot \sum_{i=1}^{n} y_i - n + 0 = 0$$

$$\frac{1}{\lambda} = \frac{n}{\sum_{i=1}^{n} y_i}, \quad \lambda = \frac{\sum_{i=1}^{n} y_i}{n} = \bar{y}$$

$$\therefore \hat{\lambda}_{MLE} = \bar{y}.$$

Example 6: $Y_1,\ldots,Y_n \overset{ind.}{\sim} Exponential(\theta)$.

$$f_Y(y) = \begin{cases} \frac{1}{\theta} e^{\frac{-y_i}{\theta}}, & y>0 \\ 0, & \text{else} \end{cases}$$

$$L(y_1 \ldots y_n | \theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-y_i/\theta}$$

$$= \log\left(\prod_{i=1}^{n} \frac{1}{\theta} e^{-y_i/\theta}\right)$$

$$= \sum_{i=1}^{n} \log\left(\frac{1}{\theta} e^{-y_i/\theta}\right)$$

$$= \sum_{i=1}^{n} \log\left(\frac{1}{\theta}\right) + \log\left(e^{-y_i/\theta}\right)$$

$$= \sum_{i=1}^{n}\left(-\log(\theta) - \frac{y_i}{\theta}\right)$$

$$= -n\log n - \frac{1}{\theta}\sum_{i=1}^{n} \log(y_i)$$

<u>Derivative</u>: $\frac{-n}{\theta} - (-\theta^{-2}) \sum_{i=1}^{n} y_i = 0$

$$\theta^{-2} \cdot \sum_{i=1}^{n} y_i = \frac{n}{\theta}$$

$$\theta = \frac{1}{n}\sum_{i=1}^{n} y_i = \bar{Y}$$

$$\therefore \bar{Y} = \theta.$$

<u>Example 7:</u> $Y_1, \ldots, Y_n \overset{ind.}{\sim} Normal(\mu_Y, \sigma_Y^2).$

<u>Example 5:</u> $X_1, \ldots, X_n \overset{ind.}{\sim} Uniform(0, \theta).$